

搭建开发环境

安装VMWare

VMWare安装包位置: [01_开发板资料包/01_开发工具/VMware-player-full-17.5.0-22583795.exe](#)

安装MobaXterm

MobaXterm安装包位置: [01_开发板资料包/01_开发工具/MobaXterm_Installer_v24.4.zip](#)

启动虚拟机

虚拟机位置: [01_开发板资料包/02_开发环境/100ASK_LVGL_Course_Ubuntu_18.04.6_VM.7z](#)

用户名: Ubuntu

密码: ubuntu

编译SDK

SDK位置: [01_开发板资料包/02_开发环境/T113s4_Lvgl9-Class_TinaSDK-V1.tar.gz](#)

- 编译SDK过程:

- 初始化环境变量

```
cd ~/100ask/T113s4_Lvgl9-Class_TinaSDK-V1
source build/envsetup.sh
```

- 执行编译:

```
ubuntu@ubuntu1804:~/100ask/T113s4_Lvgl9-Class_TinaSDK-V1$ ./build.sh
All available platform:
  0. android
  1. linux
Choice [android]: 1
All available linux_dev:
  0. bsp
  1. openwrt
Choice [bsp]: 1
All available ic:
  0. t113_s4
Choice [t113_s4]: 0
All available board:
  0. 100ask_lvgl9
Choice [100ask_lvgl9]: 0
```

```
All available flash:
  0. default
  1. nor
Choice [default]: 0
INFO: kernel relative recovery defconfig:
../../../../../device/config/chips/t113s4/configs/100ask_lvgl9/linux-5.4/config-5.4-recovery
```

3. 第一次编译会失败，这个过程必须要经历，然后执行：

```
INFO: build brandy OK.
INFO: build kernel ...
INFO: prepare_buildserver
INFO: Prepare toolchain ...
Makefile:681: arch//Makefile: No such file or directory
make: *** No rule to make target 'arch//Makefile'. Stop.
ERROR: build Failed
INFO: build kernel failed
ubuntu@ubuntu1804:~/100ask/T113s4_Lvgl9-Class_TinaSDK-V1$
ubuntu@ubuntu1804:~/100ask/T113s4_Lvgl9-Class_TinaSDK-V1$ ./build.sh -d
```

4. 编译成功的信息：

```
make[3] -C /home/ubuntu/100ask/T113s4_Lvgl9-Class_TinaSDK-V1/openwrt/target/t113_s4 install
make[2] package/index
make[2] json_overview_image_info
make[2] checksum
make: Leaving directory '/home/ubuntu/100ask/T113s4_Lvgl9-Class_TinaSDK-V1/openwrt/openwrt'
INFO: -----
INFO: build Tina OK.
INFO: -----
ubuntu@ubuntu1804:~/100ask/T113s4_Lvgl9-Class_TinaSDK-V1$
```

5. 打包得出固件：

```
ubuntu@ubuntu1804:~/100ask/T113s4_Lvgl9-Class_TinaSDK-V1$ ./build.sh pack

Add partition very amp_dsp0.fex AMP_DSP0_FEX0000
FilePath: amp_dsp0.fex
FileLength=4ecb0BuildImg 0
Dragon execute image.cfg SUCCESS !
-----image is at-----

56M /home/ubuntu/100ask/T113s4_Lvgl9-Class_TinaSDK-V1/out/t113_s4_linux_100ask_lvgl9_uart0.img

pack finish
ubuntu@ubuntu1804:~/100ask/T113s4_Lvgl9-Class_TinaSDK-V1$
```

6. 获取工具链位置：

```
cat /home/ubuntu/100ask/T113s4_Lvgl9-Class_TinaSDK-  
V1/openwrt/target/t113_s4/t113_s4-100ask_lvgl9/defconfig | grep  
"CONFIG_TOOLCHAIN"
```

从而得出工具链位置为：`/home/ubuntu/100ask/T113s4_Lvgl9-Class_TinaSDK-V1/prebuilt/rootfsbuilt/arm/toolchain-sunxi-glibc-gcc-830/`

配置工具链环境变量：

```
export PATH=$PATH:/home/ubuntu/100ask/T113s4_Lvgl9-Class_TinaSDK-  
V1/prebuilt/rootfsbuilt/arm/toolchain-sunxi-glibc-gcc-830/toolchain/bin  
export CROSS_COMPILE=arm-openwrt-linux-gnueabi  
export STAGING_DIR=/home/ubuntu/100ask/T113s4_Lvgl9-Class_TinaSDK-  
V1/out/t113_s4/100ask_lvgl9/openwrt/staging_dir/target  
  
# STAGING_DIR 在OpenWrt工具链中用于管理构建过程中的中间文件，确保构建环境的隔离和交叉编译的顺利进行。
```

- 如果想重新编译怎么操作：

```
rm -rf out/ .buildconfig
```

之后，从头开始上面的操作

技术交流学习

欢迎加入讨论：

- 社区交流：<https://forums.100ask.net>
- QQ技术交流群（如群满，请加qq：401684796 验证备注：LVGL）：962138234
- 微信交流群：添加微信：baiwenkeji_fae 验证备注：LVGL